

Module/Pathway/Taxon Review:

Glutathione-dependent Formaldehyde Detoxification in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00680` "Methane metabolism" (module area: energy_respiration_inorganic_metabolism) **Module under review:** Glutathione-dependent formaldehyde detoxification (chemical capture → hemithioacetal oxidation → thioester hydrolysis)

1. Executive Summary

The glutathione (GSH)-dependent formaldehyde detoxification module is **present and satisfiable** in *Pseudomonas putida* KT2440, but with two curation-critical qualifications. First, the module's enzymatic core reduces to exactly **two adjacent, single-copy genes**: `frmA` (**PP_1616**, Q88MF5; S-(hydroxymethyl)glutathione dehydrogenase, EC 1.1.1.1 / 1.1.1.284, KO K00121) and `frmC` (**PP_1617**, Q88MF4; S-formylglutathione hydrolase, EC 3.1.2.12, KO K01070). These form a compact two-gene operon (9-bp intergenic gap) controlled by a divergently transcribed LysR-type regulator, PP_1615. Step 1 of the module — capture of formaldehyde by glutathione to form S-(hydroxymethyl)glutathione — is a **spontaneous, non-enzymatic** chemical reaction in this organism; no formaldehyde-activating enzyme (Gfa) is encoded in the proteome, which is expected because KT2440 is not a methylotroph.

Second, and most important for module satisfiability, the GSH route is a **secondary/backup pathway** in KT2440. The organism detoxifies formaldehyde predominantly through **glutathione-INDEPENDENT** NAD-linked formaldehyde dehydrogenases — `fdhA`/PP_0328 (EC 1.2.1.46, KO K00148, carrying the diagnostic InterPro IPR014184 "HCHO_DH_non_GSH" signature) and PP_3970 (a Zn/NAD(P) oxidoreductase not in the candidate list). Roca et al. (2009) showed that a double mutant lacking both GSH-independent enzymes still evolved CO₂

from formaldehyde, implying an additional route — plausibly the GSH-dependent FrmA pathway acting as backup. The resulting formate is oxidized to CO₂ by two formate dehydrogenase clusters (PP_0489–0492 and PP_2183–2186).

The dominant curation problem is that the supplied 30-gene candidate list is an artifact of **KEGG map-level (ppu00680 "Methane metabolism") membership**, not true module membership. Only `frmA` and `frmC` belong to the GSH-dependent formaldehyde-oxidation steps. The remaining ~28 genes are serine-cycle / serine-biosynthesis enzymes, downstream formate dehydrogenases, central-metabolism enzymes, and an amine dehydrogenase that co-occur on the broad overview map. **Recommendation:** mark steps 2 and 3 covered by `frmA` and `frmC` respectively, mark step 1 covered (*non-enzymatic*), flag `frmA` and `frmC` for full `fetch-gene` review, and **revise the bucket** into a dedicated {frmA, frmC} module distinct from the KEGG methane-metabolism overview map.

2. Target-Organism Pathway Definition

What this module includes

The GSH-dependent formaldehyde detoxification module is a three-step process that converts cytotoxic formaldehyde into formate while regenerating glutathione:

Step	Reaction	Catalyst in KT2440	Nature
1. GSH capture	Formaldehyde + GSH → S-(hydroxymethyl)glutathione (hemithioacetal)	none (spontaneous)	Non-enzymatic
2. Hemithioacetal oxidation	S-(hydroxymethyl)glutathione + NAD ⁺ → S-formylglutathione + NADH + H ⁺	FrmA / PP_1616	Enzymatic
3. Thioester hydrolysis	S-formylglutathione + H ₂ O → formate + GSH + H ⁺	FrmC / PP_1617	Enzymatic

The output, **formate**, is then handed off to formate dehydrogenase(s) for oxidation to CO₂ — but that oxidation is a **separate downstream pathway**, not part of the detox module proper.

Neighboring pathways to keep separate

For curation, the following processes are represented in the candidate list because they share the KEGG ppu00680 overview map, but must be kept **outside** the GSH-detox module boundary:

- **Glutathione-independent formaldehyde oxidation** (`fdhA` /PP_0328, PP_3970) — a *parallel* detox route, mechanistically and evolutionarily distinct (different KO: K00148 vs. K00121).
- **Formate oxidation to CO₂** (`fdoGHI` /PP_0489–0491; PP_2183–2186) — downstream of the module.
- **Serine cycle / C1 assimilation and serine biosynthesis** (`glyA1` , `glyA2` , `serA` , `serB` , `serC` , `hprA` , `gpmI` , `eno` , `ppc` , `ppsA` , `fba` , `fbp` , `mdh` , `pta` , `ttuD` , PP_2533).
- **Acetate/acyl-CoA activation** (`acsA1` , `acsA2` , PP_2213).
- **Amine oxidation** (`peaA` , quinohaemoprotein amine dehydrogenase).

Alternate names / database definitions

- The FrmA enzyme has multiple synonyms: **class-III (class-3) alcohol dehydrogenase, glutathione-dependent formaldehyde dehydrogenase, S-(hydroxymethyl)glutathione dehydrogenase**, ADH5 (eukaryotic ortholog). EC numbers 1.1.1.1 and 1.1.1.284 both apply.
 - FrmC = **S-formylglutathione hydrolase (FGH)**, EstD in humans, EC 3.1.2.12.
 - In enterobacteria the operon is `frmRAB` (regulator–dehydrogenase–hydrolase); KT2440 lacks the enterobacterial FrmR sensor and uses a LysR-type regulator instead (see §5).
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3. Expected Step Model and Satisfiability

```

Formaldehyde (HCHO)
|
| + GSH (STEP 1: spontaneous chemical capture – NON-ENZYMATIC)
v
S-(hydroxymethyl)glutathione [hemithioacetal]
|
| NAD+ --> NADH (STEP 2: FrmA / PP_1616, EC 1.1.1.284, K00121) <-- COVERED
v
S-formylglutathione [thioester]
|
| + H2O (STEP 3: FrmC / PP_1617, EC 3.1.2.12, K01070) <-- COVERED
v
Formate + GSH (regenerated)
|
| (downstream, SEPARATE pathway: formate dehydrogenases
| PP_0489-0492, PP_2183-2186)
v
CO2

```

Satisfiability verdict:

Module step	Status	Encoded by	Confidence
1. GSH capture of formaldehyde	covered (non-enzymatic)	none required	High — reaction is spontaneous; module outline itself labels it "chemical capture"
2. Hemithioacetal oxidation	covered	<code>frmA</code> / PP_1616	High — direct UniProt annotation + operon context + literature
3. S-formylglutathione hydrolysis	covered	<code>frmC</code> / PP_1617	High — direct UniProt annotation + operon context

No expected step is a *gap*. No step requires a lineage-specific replacement. The only "missing" component relative to the enterobacterial paradigm is the FrmR regulator, and its absence is expected (see §5) and does not affect catalytic satisfiability.

4. Candidate Genes and Evidence

4.1 High-confidence module core

frmA — PP_1616 (Q88MF5), S-(hydroxymethyl)glutathione dehydrogenase

FrmA is a zinc-containing, class-III alcohol dehydrogenase that catalyzes the NAD⁺-dependent oxidation of the GSH-formaldehyde hemithioacetal: S-(hydroxymethyl)glutathione + NAD⁺ → S-formylglutathione + NADH + H⁺ (EC 1.1.1.1 / 1.1.1.284). UniProt Q88MF5 explicitly notes "Has high formaldehyde dehydrogenase activity in the presence of glutathione." This is the **central, defining reaction** of the module. Osman et al. (2016) confirm that "Formaldehyde detoxification by FrmA requires S-(hydroxymethyl)glutathione" [PMID: 27474740], establishing that FrmA acts on the GSH adduct rather than free formaldehyde. Evidence type for KT2440: strong homology-based annotation plus operon/genomic-context support; direct enzymology is from orthologs. **Curation caveat:** the KEGG primary bucket for **frmA** in the metadata is **ppu00626** (naphthalene/aminobenzoate — likely a class-III ADH broad mapping), which is misleading; the functionally correct assignment is the GSH-detox module. Promote to full review.

frmC — PP_1617 (Q88MF4), S-formylglutathione hydrolase

FrmC is a serine hydrolase (GXSVG-type esterase fold) that hydrolyzes the S-formylglutathione thioester: S-formylglutathione + H₂O → formate + glutathione + H⁺ (EC 3.1.2.12). UniProt Q88MF4 describes it as "Serine hydrolase involved in the detoxification of formaldehyde." This step both releases formate and **regenerates glutathione**, closing the cycle. Homology-based support is strengthened by the note that lactococcal EstA and human EstD/FGH share significant similarity as "part of a universal formaldehyde detoxification pathway" [PMID: 10742212]. Evidence type for KT2440: strong homology-based annotation + operon context. Promote to full review.

4.2 Parallel / downstream / off-module candidates (do NOT count toward module coverage)

Gene	Locus	Assigned role	Why excluded from GSH module
fdhA	PP_0328	Formaldehyde dehydrogenase, EC 1.2.1.46	Glutathione-INDEPENDENT (IPR014184 "HCHO_DH_non_GSH" KO K00148) — parallel route, not GSH module
(PP_3970)	PP_3970	Zn/NAD(P) oxidoreductase	Second GSH-independent formaldehyde dehydrogenase; not candidate list
fdoG/H/I	PP_0489–0491	Formate dehydrogenase-O	Downstream format → CO ₂ , separate pathway
PP_2183–2186	—	NDH-1 subunits + formate dehydrogenase α/δ	Downstream format oxidation / respiration
glyA1 , glyA2	PP_0322, PP_0671	Serine hydroxymethyltransferase	Serine/one-carbon metabolism (bucket ppu04981)
serA , serB , serC	PP_5155, PP_4909, PP_1768	Serine biosynthesis	Amino-acid biosynthesis, not detox
hprA , ttuD , PP_2533	PP_0762, PP_4300, PP_2533	2-hydroxyacid/hydroxypyruvate reductases	Serine cycle / C2 metabolism
eno , gpmI , fba , fbp , ppc , ppsA , mdh , pta	various	Central carbon metabolism	Glycolysis/gluconeogenesis/anaplerosis
acsA1 , acsA2 , PP_2213	PP_4487, PP_4702, PP_2213	Acetyl-CoA synthetases / acyl-CoA synthetase	Acetate activation

Gene	Locus	Assigned role	Why excluded from GSH module
peaA	PP_5602	Quinohaemoprotein amine dehydrogenase	Amine oxidation (a formaldehyde source not detox)

Of the 30 candidates, **28 are KEGG map-level associations** that co-occur on the ppu00680 "Methane metabolism" overview map but do not participate in the GSH-dependent detox reactions.

5. Gaps, Ambiguities, and Likely Over-annotations

5.1 The candidate list is a KEGG overview-map artifact

The single largest curation issue: ppu00680 "Methane metabolism" is a **broad overview map**, and 28 of 30 candidates were pulled in by map membership rather than by participation in the GSH-detox reactions (Finding F003). This inflates apparent module complexity. The true module core is frmA + frmC. Any automated satisfiability call that treats all 30 as module members would be wrong.

5.2 No paralog ambiguity for the core enzymes

KEGG KO→gene mapping in *P. putida* KT2440 is clean and single-copy (Finding F007):

- K00121 (S-(hydroxymethyl)glutathione dehydrogenase) → **PP_1616 only**
- K01070 (S-formylglutathione hydrolase) → **PP_1617 only**

There is no second copy of either enzyme to create ambiguity. The GSH-independent fdhA / PP_0328 is cleanly separated into a distinct ortholog group (K00148), so it will not be mis-assigned to the GSH module at the KO level.

5.3 No Gfa; no enterobacterial FrmR

A proteome-wide InterPro/Pfam screen (Finding F006) confirmed:

- **No bona fide Gfa** (glutathione-dependent formaldehyde-activating enzyme). The single Pfam PF04244 hit (Q88JB7 / PP_2732) is a cryptochrome/photolyase family protein, not Gfa.

This is expected: Gfa accelerates hemithioacetal formation in methylotrophs (e.g., *Paracoccus denitrificans*, *Methylobacterium*), and KT2440 is not a methylotroph. Step 1 therefore remains non-enzymatic in this organism.

- **No FrmR formaldehyde sensor.** The only DUF156-family protein (Pfam PF02583 / IPR003735) is Q88IN0 / PP_2969 = **CsoR**, a copper-sensing repressor, NOT a formaldehyde-sensing FrmR. Instead, the frm operon is controlled by a divergent **LysR-type regulator**, **PP_1615** (Q88MF6) (Findings F003, F005).

5.4 Broad EC / bucket mislabels to correct

- **frmA** primary bucket in metadata is **ppu00626**; functionally it belongs to formaldehyde detox. The class-III ADH annotation (EC 1.1.1.1) is broad; the module-relevant activity is EC 1.1.1.284.
- **frmC** metadata bucket is correctly **ppu00680**, but should be re-scoped to the dedicated detox module.
- **fdhA** is at risk of being conflated with GSH-dependent formaldehyde dehydrogenase because both share the phrase "formaldehyde dehydrogenase"; the IPR014184 signature and EC 1.2.1.46 firmly place it in the **GSH-independent** class.

5.5 Genomic context of the operon

KEGG coordinates (Finding F005): **frmA** PP_1616 at 1,812,522–1,813,637 (+); **frmC** PP_1617 at 1,813,646–1,814,500 (+) — a 9-bp intergenic gap consistent with a single transcriptional unit. PP_1615 (LysR regulator) sits divergently on the (–) strand at 1,811,521–1,812,417, ~105 bp upstream of **frmA**. Flanking genes PP_1614 (**ispD**) and PP_1618 (**ispF**) are MEP-pathway isoprenoid genes, unrelated to formaldehyde metabolism — so the operon is compact and cleanly bounded.

(-) strand		(+) strand ----->				
PP_1614	PP_1615		PP_1616 (frmA)	--9bp--	PP_1617 (frmC)	PP_1618
ispD	LysR reg	<-->	S-(hydroxymethyl)GSH		S-formylGSH	ispF
(MEP)	(divergent)		dehydrogenase		hydrolase	(MEP)

6. Module and GO-curation Recommendations

6.1 Step-level status calls

Module step	Recommended status	Rationale
1. GSH capture of formaldehyde	covered — non-enzymatic	Spontaneous hemithioacetal formation; no gene needed; Gfa correctly <i>not_expected_in_target_taxon</i>
2. Hemithioacetal oxidation	covered	<code>frmA</code> / PP_1616 (K00121), single-copy, high confidence
3. S-formylglutathione hydrolysis + GSH regeneration	covered	<code>frmC</code> / PP_1617 (K01070), single-copy, high confidence

No step is `gap` or `candidate_uncertain` at the catalytic level.

6.2 Bucket / module boundary corrections (`module_needs_revision`)

- **The bucket needs revision.** The generic `ppu00680` "Methane metabolism" bucket is the wrong container for this module in KT2440. Create/point to a **dedicated {frmA, frmC} GSH-dependent formaldehyde detoxification** module, explicitly excluding the 28 overview-map co-members.
- Mark the parallel **GSH-independent** detox route (`fdhA` /PP_0328, PP_3970) as a *separate module* — it is biologically the dominant route and should be tracked, but not folded into the GSH module.
- Optionally annotate a *downstream* formate-oxidation module (PP_0489–0492, PP_2183–2186) as adjacent but distinct.

6.3 Species-aware note for module satisfiability

The GSH module is **satisfiable but non-essential** in KT2440. Because the organism carries redundant GSH-independent formaldehyde dehydrogenases (Finding F001), a satisfiability engine should record the GSH route as a **backup/secondary** pathway. This context prevents an over-interpretation that FrmA/FrmC are the primary formaldehyde-clearance machinery.

6.4 GO-term requests

No new GO terms appear necessary. Existing terms cover the reactions: GO:0004022/GO:0051903 (S-(hydroxymethyl)glutathione dehydrogenase / alcohol dehydrogenase [NAD⁺] activity), GO:0018738 (S-formylglutathione hydrolase activity), and GO:0046294 (formaldehyde catabolic process). Ensure `frmA` carries the class-III/GSH-dependent activity annotation (GO:0051903) and not merely generic alcohol dehydrogenase; and that `fdhA` is annotated to the **glutathione-independent** formaldehyde dehydrogenase activity to keep the two routes distinct.

7. Genes to Promote to Full `fetch-gene` Review

Gene	Locus	UniProt	Priority	Reason
<code>frmA</code>	PP_1616	Q88MF5	High	Core module step 2; correct EC 1.1.1.284 vs broad EC 1.1.1.1; fix misleading ppu00626 bucket
<code>frmC</code>	PP_1617	Q88MF4	High	Core module step 3; confirm S-formylglutathione hydrolase and GSH regeneration
<code>fdhA</code>	PP_0328	Q88R06	Medium	Ensure annotation as GSH-independent (K00148, IPR014184); prevent conflation with FrmA
PP_1615	PP_1615	Q88MF6	Medium	Confirm LysR-type regulator of frm operon; document divergent transcription
PP_3970	PP_3970	Q88FV8	Low	Second GSH-independent formaldehyde dehydrogenase; add to parallel module if tracked

8. Mechanistic Model / Interpretation

P. putida KT2440 possesses a **layered formaldehyde defense**. The frontline is a set of **glutathione-independent, NAD-linked formaldehyde dehydrogenases** (`fdhA`/PP_0328 and PP_3970) that directly oxidize free formaldehyde to formate. Roca et al. (2009) demonstrated this redundancy: single and double knockouts of the two GSH-independent enzymes both

retained the ability to evolve $^{14}\text{CO}_2$ from ^{14}C -formaldehyde, and a double mutant *still* evolved CO_2 , implying at least a third route [PMID: 19304846]. That residual activity is most parsimoniously explained by the **GSH-dependent FrmA/FrmC pathway acting as a backup**.

In the GSH route, formaldehyde is first captured spontaneously by cellular glutathione to form the hemithioacetal S-(hydroxymethyl)glutathione — no enzyme (Gfa) is needed or encoded, consistent with KT2440's non-methylotrophic lifestyle. FrmA (PP_1616) then oxidizes the hemithioacetal to the thioester S-formylglutathione using NAD^+ , and FrmC (PP_1617) hydrolyzes the thioester to release formate and regenerate glutathione. Both enzymatic steps are encoded by a tight two-gene operon under a dedicated LysR-type regulator (PP_1615). All formaldehyde-derived formate — from either route — converges on the cell's **formate dehydrogenase clusters** (PP_0489–0492 and PP_2183–2186) for terminal oxidation to CO_2 , allowing near-stoichiometric conversion of formaldehyde to CO_2 (physiological study, [PMID: 21261833]).

This layered model has a clear curation implication: the *molecular identity* of the GSH module is minimal and unambiguous (two single-copy genes), while its *physiological weight* is secondary. The candidate list's apparent breadth is an accident of KEGG overview-map bookkeeping, not biology.

9. Evidence Base

PMID	Title (abbrev.)	How it supports/challenges the review
PMID: 19304846	<i>Redundancy of enzymes for formaldehyde detoxification in P. putida</i>	Core evidence that KT2440 uses redundant GSH-INDEPENDENT formaldehyde dehydrogenases as the dominant route; residual CO ₂ evolution in the double mutant implies an additional (GSH) route. Establishes the "backup" status of the module.
PMID: 21261833	<i>Physiological responses of P. putida to formaldehyde</i>	Confirms two formaldehyde dehydrogenases + two formate dehydrogenase complexes stoichiometrically convert formaldehyde to CO ₂ ; glutathione-biosynthesis mutants fail to grow at 1.5 mM HCHO, supporting a role for the GSH system.
PMID: 27474740	<i>Effectors and sensory sites of FrmR</i>	Confirms "Formaldehyde detoxification by FrmA requires S-(hydroxymethyl)glutathione" — anchors FrmA's substrate as the GSH adduct (module step 2).
PMID: 16242864	<i>Coupling of vanillate-O-demethylase and formaldehyde detox</i>	Shows formaldehyde (a demethylation by-product) is converted to formate, and that <i>frmA</i> (GSH-dependent formaldehyde dehydrogenase) disruption blocks formate accumulation — supports FrmA's detox function (in a coupled <i>P. putida</i> / <i>E. coli</i> context).
PMID: 11133961	<i>Two-component regulation of methanol/formaldehyde oxidation in Paracoccus denitrificans</i>	Documents the GSH-dependent formaldehyde dehydrogenase (fhlA) + S-formylglutathione hydrolase (fghA) pair in a methylotroph — the context where Gfa-type activation is relevant, contrasting with non-methylotroph KT2440.
PMID: 10742212	<i>Tributylin esterase of Lactococcus lactis</i>	Establishes S-formylglutathione hydrolase (FGH) as part of a "universal formaldehyde detoxification pathway" via sequence similarity — supports FrmC family assignment.
PMID: 37273222	<i>Core and auxiliary functions of one-carbon metabolism</i>	Context for one-carbon/serine metabolism that populates much of the ppu00680 overview map — supports excluding those genes from the detox module.

Direct experimental evidence for the target strain: the redundancy and physiology of formaldehyde detoxification (PMIDs 19304846, 21261833) are from *P. putida* KT2440 itself. **Homology/database-inferred evidence:** the specific catalytic assignments of FrmA and FrmC in KT2440 rest on UniProt annotation and operon context, with direct enzymology drawn from orthologs (Paracoccus, human, lactococcal FGH).

10. Limitations and Knowledge Gaps

1. **No direct KT2440 enzymology for FrmA/FrmC.** The catalytic assignments are annotation- and homology-based (UniProt Q88MF5, Q88MF4). No published in-vitro kinetics or knockout-phenotype paper isolates FrmA/FrmC activity in KT2440 specifically. The "backup route" interpretation is inferred from residual activity in GSH-independent double mutants (P 19304846), not from a direct frmA/frmC knockout.
2. **Regulator not experimentally validated.** PP_1615 is annotated as LysR-type and is positioned divergently, but its role as the frm-operon regulator and its inducer (formaldehyde? S-formylglutathione?) have not been experimentally demonstrated in KT2440.
3. **Step 1 kinetics.** The assumption that hemithioacetal formation is fully non-enzymatic is standard for non-methylotrophs, but glutathione pool size and any accessory factors under formaldehyde stress in KT2440 are not quantified here.
4. **PP_3970 not in candidate list.** Identified from proteome screening; its exact contribution relative to fdhA is not resolved.
5. **Literature scope.** Two of the seven retrieved papers (Paracoccus, Lactococcus) are non-target organisms used for cross-species inference; transfer of mechanistic detail to KT2440 is homology-based, not direct.

11. Proposed Follow-up Experiments / Actions

Curation actions (immediate): 1. Split the `ppu00680` bucket: create a dedicated **{frmA, frmC} GSH-dependent formaldehyde detoxification** module; move the 28 overview-map co-members out. 2. Create a separate **GSH-independent formaldehyde oxidation** module ({fdhA/PP_0328, PP_3970}) and a **formate → CO₂** module (fdo clusters). 3. Correct `frmA` bucket (from ppu00626) and ensure EC 1.1.1.284 / GO:0051903 are primary. 4. Promote `frmA`, `frmC` (and secondarily `fdhA`, PP_1615) to full `fetch-gene` review.

Wet-lab / expert questions (to resolve gaps): 5. Construct single and double `ΔfrmA / ΔfrmC` mutants (and combine with `ΔfdhA ΔPP_3970`) and measure formaldehyde tolerance and ¹⁴CO₂ evolution to directly quantify the GSH route's backup contribution. 6. Test whether PP_1615 deletion derepresses or abolishes frmA-frmC expression, and identify the inducing signal. 7. Verify the frmA-frmC operon by RT-PCR / RNA-seq (predicted single transcript from the 9-bp gap). 8. Confirm in-vitro that KT2440 FrmA requires the GSH-formaldehyde adduct (not free formaldehyde) and that FrmC regenerates GSH.

Key References

1. Roca A, Rodríguez-Herva JJ, Ramos JL. *Redundancy of enzymes for formaldehyde detoxification in Pseudomonas putida*. PMID: 19304846
2. Roca A et al. *Physiological responses of Pseudomonas putida to formaldehyde during detoxification*. PMID: 21261833
3. Osman D et al. *The effectors and sensory sites of formaldehyde-responsive regulator FrmR and metal-sensing variant*. PMID: 27474740
4. *Functional coupling between vanillate-O-demethylase and formaldehyde detoxification pathway*. PMID: 16242864
5. Harms N et al. *Two-component system that regulates methanol and formaldehyde oxidation in Paracoccus denitrificans*. PMID: 11133961
6. Fernández de Palencia P et al. *Cloning and characterization of the major tributyrin esterase (S-formylglutathione hydrolase homolog) of Lactococcus lactis*. PMID: 10742212
7. *Core and auxiliary functions of one-carbon metabolism*. PMID: 37273222

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